General Relativity

Final Examination — Answers

Question 1. *(2 points)*
Transformation law. Under a change of coordinates $x^\mu \to x'^\mu(x)$ the Riemann tensor, a $(1,3)$ tensor, transforms as

$$R'^{\mu}{}_{\nu\rho\sigma} = \frac{\partial x'^\mu}{\partial x^\alpha}\,\frac{\partial x^\beta}{\partial x'^\nu}\,\frac{\partial x^\gamma}{\partial x'^\rho}\,\frac{\partial x^\delta}{\partial x'^\sigma}\; R^\alpha{}_{\beta\gamma\delta}\,. \tag{1}$$

Ricci tensor. By definition $R'_{\mu\nu} = R'^{\rho}{}_{\mu\rho\nu}$. Inserting the law above (upper index $\to \rho$, and the third index set equal to ρ):

$$R'_{\mu\nu} = R'^{\rho}{}_{\mu\rho\nu} = \frac{\partial x'^\rho}{\partial x^\alpha}\,\frac{\partial x^\beta}{\partial x'^\mu}\,\frac{\partial x^\gamma}{\partial x'^\rho}\,\frac{\partial x^\delta}{\partial x'^\nu}\; R^\alpha{}_{\beta\gamma\delta}\,. \tag{2}$$

The sum over ρ involves only the two Jacobians $\frac{\partial x'^\rho}{\partial x^\alpha}\frac{\partial x^\gamma}{\partial x'^\rho}$, and by the chain rule

$$\frac{\partial x^\gamma}{\partial x'^\rho}\frac{\partial x'^\rho}{\partial x^\alpha} = \frac{\partial x^\gamma}{\partial x^\alpha} = \delta^\gamma_\alpha\,. \tag{3}$$

Therefore

$$R'_{\mu\nu} = \frac{\partial x^\beta}{\partial x'^\mu}\,\frac{\partial x^\delta}{\partial x'^\nu}\;\delta^\gamma_\alpha R^\alpha{}_{\beta\gamma\delta} = \frac{\partial x^\beta}{\partial x'^\mu}\,\frac{\partial x^\delta}{\partial x'^\nu}\; R^\alpha{}_{\beta\alpha\delta} = \frac{\partial x^\beta}{\partial x'^\mu}\,\frac{\partial x^\delta}{\partial x'^\nu}\; R_{\beta\delta}\,, \tag{4}$$

which is precisely the transformation law of a $(0,2)$ tensor. ■

Question 2. *(3 points)*
Write $\hat\Gamma_{\mu\nu}{}^\rho = \Gamma_{\mu\nu}{}^\rho + K_{\mu\nu}{}^\rho$ with $K_{\mu\nu}{}^\rho = \frac{1}{12} g^{\rho\sigma} H_{\mu\nu\sigma}$, so that $K_{\mu\nu\sigma} = \frac{1}{12} H_{\mu\nu\sigma}$ is totally antisymmetric. (The difference of two connections is a tensor, so this is a legitimate connection.)

a) **Torsion.** The Levi-Civita connection is symmetric, $\Gamma_{\mu\nu}{}^\rho = \Gamma_{\nu\mu}{}^\rho$, so the torsion is

$$\hat T_{\mu\nu}{}^\rho = \hat\Gamma_{\mu\nu}{}^\rho - \hat\Gamma_{\nu\mu}{}^\rho = K_{\mu\nu}{}^\rho - K_{\nu\mu}{}^\rho = \frac{1}{12} g^{\rho\sigma}\left(H_{\mu\nu\sigma} - H_{\nu\mu\sigma}\right) = \frac{1}{6}\, g^{\rho\sigma} H_{\mu\nu\sigma}\,. \tag{5}$$

This is non-zero whenever $H \neq 0$, so $\hat\Gamma$ is *not* torsion-free (it is torsion-free only if $H = 0$).

Metric compatibility. Since the Levi-Civita part satisfies $\nabla_\sigma g_{\mu\nu} = 0$,

$$\hat\nabla_\sigma g_{\mu\nu} = \nabla_\sigma g_{\mu\nu} - K_{\sigma\mu}{}^\lambda g_{\lambda\nu} - K_{\sigma\nu}{}^\lambda g_{\mu\lambda} = -K_{\sigma\mu\nu} - K_{\sigma\nu\mu} = -\frac{1}{12}\left(H_{\sigma\mu\nu} + H_{\sigma\nu\mu}\right) = 0\,, \tag{6}$$

by the antisymmetry of H in its last two indices. (Because H is totally antisymmetric the result is the same whichever of the first two indices of $\hat\Gamma$ is taken as the derivative index.) So $\hat\Gamma$ *is* metric compatible.

b) The geodesic equation for $\hat{\Gamma}$ is

$$\frac{d^2x^\rho}{d\tau^2} + \hat{\Gamma}_{\mu\nu}{}^\rho \frac{dx^\mu}{d\tau}\frac{dx^\nu}{d\tau} = 0 \quad \Longleftrightarrow \quad \frac{d^2x^\rho}{d\tau^2} + \Gamma_{\mu\nu}{}^\rho \frac{dx^\mu}{d\tau}\frac{dx^\nu}{d\tau} + \frac{1}{12} g^{\rho\sigma} H_{\mu\nu\sigma} \frac{dx^\mu}{d\tau}\frac{dx^\nu}{d\tau} = 0\,. \quad (7)$$

The last term contracts the symmetric object $\dot{x}^\mu \dot{x}^\nu$ with $H_{\mu\nu\sigma}$, which is antisymmetric in $\mu\nu$, so it vanishes identically. The equation reduces to the Levi-Civita geodesic equation (with the same affine parameter), so the geodesics coincide. (Only the symmetric part of a connection matters for geodesics; the torsion drops out.)

Question 3. *(4 points)*

a) The metric components are singular at $r = 0$, at $r = 2GM$ and (trivially) at $\theta = 0, \pi$. The latter are just the usual coordinate singularities of spherical polar coordinates (the same as the poles of a globe), removed by going to Cartesian-like coordinates.

- $r = 2GM$ (the event horizon), where $g_{tt} \to 0$ and $g_{rr} \to \infty$, is a *coordinate singularity*: it is a failure of the (t, r) coordinates, not of the geometry. All curvature invariants are finite there; e.g. the Kretschmann scalar is

$$R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} = \frac{48\,G^2M^2}{r^6} \xrightarrow{r=2GM} \frac{3}{4G^4M^4} < \infty, \quad (8)$$

  and a freely falling observer crosses it in finite proper time and feels nothing special (tidal forces $\sim GM/r^3$ are finite and small for a large black hole).
- $r = 0$ is a genuine *curvature singularity*: $R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} = 48G^2M^2/r^6 \to \infty$, which is a coordinate-independent statement. Tidal forces diverge, and every infalling observer reaches it in finite proper time (at most πGM for radial infall from the horizon) where geodesics end. This is what one should be concerned about.

(The horizon is still one-way: once inside, one cannot return, but that is not a singularity of the metric.)

b) Define $r_* = r + 2GM \ln|r/2GM - 1|$, so $v = t + r_*$. Then

$$\frac{dr_*}{dr} = 1 + \frac{2GM}{r - 2GM} = \frac{r}{r - 2GM} = \frac{1}{f(r)}\,, \qquad dv = dt + \frac{dr}{f} \;\Rightarrow\; dt = dv - \frac{dr}{f}\,. \quad (9)$$

Substituting,

$$-f\,dt^2 = -f\left(dv^2 - \frac{2}{f}dv\,dr + \frac{dr^2}{f^2}\right) = -f\,dv^2 + 2\,dv\,dr - \frac{dr^2}{f}\,, \quad (10)$$

and adding $f^{-1}dr^2$ the dr^2 terms cancel:

$$ds^2 = -\left(1 - \frac{2GM}{r}\right) dv^2 + 2\, dv\, dr + r^2 \left(d\theta^2 + \sin^2\theta\, d\phi^2\right). \tag{11}$$

Interpretation. In the (v, r) block the metric is $\begin{pmatrix} -f & 1 \\ 1 & 0 \end{pmatrix}$ with determinant -1: it is finite, smooth and invertible at $r = 2GM$ (all components are analytic there, and the inverse metric is regular too). Thus $r = 2GM$ is only a coordinate artefact of (t, r), and the spacetime extends smoothly through it into $r < 2GM$. In these coordinates the radial null rays satisfy $dv(-f\, dv + 2\, dr) = 0$: ingoing rays are $v = \text{const}$, and the other family has $dr/dv = f/2$, which for $r < 2GM$ is negative, so that *all* future light cones point towards decreasing r and nothing escapes. The only place where the metric genuinely misbehaves is $r = 0$ ($f \to \infty$), consistent with the curvature singularity found in a). Hence one can fall through $r = 2GM$ safely but not avoid $r = 0$.

Question 4. *(5 points)*

a) The metric has the Friedmann–Lemaître–Robertson–Walker form

$$ds^2 = -dt^2 + a(t)^2\, d\Omega_3^2\,, \qquad d\Omega_3^2 = d\chi^2 + \sin^2\chi\, d\Omega_2^2\,, \tag{12}$$

with spatial slices that are round three-spheres ($k = +1$). Spatial slices are homogeneous and isotropic, as the cosmological principle and the observed large-scale uniformity (and the CMB isotropy) require. The **scale factor** is

$$a(t) = L\cosh(t/L)\,. \tag{13}$$

It is a solution of the Einstein equations with a positive cosmological constant: with $\dot{a} = \sinh(t/L)$,

$$H^2 + \frac{k}{a^2} = \frac{\dot{a}^2 + 1}{a^2} = \frac{\cosh^2(t/L)}{L^2\cosh^2(t/L)} = \frac{1}{L^2} = \frac{\Lambda}{3}\,, \qquad \Lambda = \frac{3}{L^2}, \tag{14}$$

which is the Friedmann equation for vacuum energy only. At late times $a \simeq \frac{L}{2}e^{t/L}$: exponential, accelerating expansion, as seen in our universe at present (dark energy) and as in inflation. It is a good model of a Λ-dominated universe, although an exactly empty one is only an approximation to ours.

b) Suppose $g_{\mu\nu} = \Omega^2 \bar{g}_{\mu\nu}$ with Ω smooth and nowhere zero. For any vector V, $g(V, V) = \Omega^2\, \bar{g}(V, V)$, so $g(V, V)$ and $\bar{g}(V, V)$ have the same sign: timelike, null and spacelike vectors and curves are the same in both metrics. In particular, the light cones are

identical (null curves $ds^2 = 0$ coincide, $\Omega^2 \bar{g} = 0 \Leftrightarrow \bar{g} = 0$), and so are the notions of causal curves, time orientation, and causal past and future. Causal structure depends only on the light-cone structure, hence is the same. (Distances and proper times differ, but they are not part of the causal structure.)

c) With $\cosh(t/L) = 1/\cos\eta$ and $t \in (-\infty, \infty)$ we get $\eta \in (-\pi/2, \pi/2)$, and $\sinh(t/L) = \sqrt{\cosh^2 - 1} = \tan\eta$ (same sign as t). Differentiating,

$$\frac{1}{L}\sinh(t/L)\,dt = \frac{\sin\eta}{\cos^2\eta}\,d\eta \;\Rightarrow\; dt = \frac{L\,d\eta}{\cos\eta}\,. \tag{15}$$

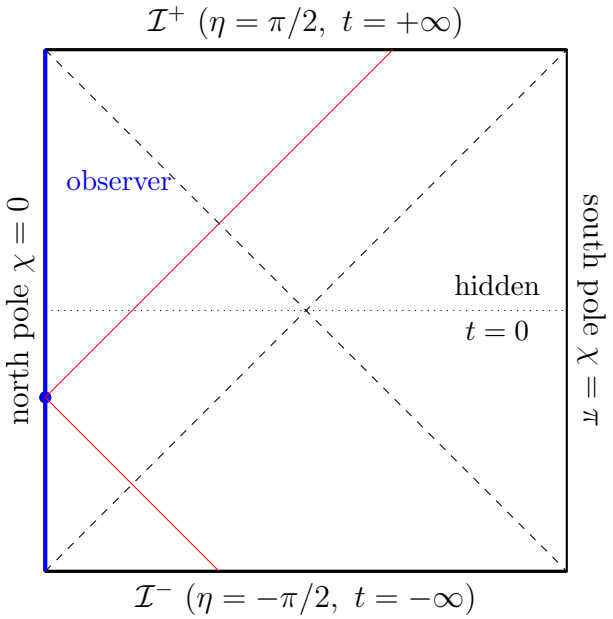
Therefore

$$ds^2 = \frac{L^2}{\cos^2\eta}\left[-d\eta^2 + d\chi^2 + \sin^2\chi\left(d\theta^2 + \sin^2\theta\,d\phi^2\right)\right], \tag{16}$$

which is conformal (with $\Omega = L/\cos\eta$) to the Einstein static universe $\mathbb{R}\times S^3$ restricted to the finite strip $-\pi/2 < \eta < \pi/2$, $0 \le \chi \le \pi$. By b), de Sitter therefore has the same causal structure as this finite region of the Einstein static universe, and radial null rays obey $d\eta = \pm d\chi$, i.e. are straight lines at 45° in the (χ, η) plane.

Penrose diagram. It is a square of width π (in χ) and height π (in η), each point representing a two-sphere of radius $L\sin\chi/\cos\eta$:

- the left edge $\chi = 0$ and the right edge $\chi = \pi$ are the north and south poles of the S^3: these are *not* boundaries but lines of origin, along which the sphere shrinks to zero size (timelike worldlines of observers at the poles);
- the top edge $\eta = \pi/2$ is future infinity $\mathcal{I}^+$ ($t \to \infty$) and the bottom edge $\eta = -\pi/2$ is past infinity $\mathcal{I}^-$ ($t \to -\infty$); both are *spacelike*;
- the horizontal line $\eta = 0$ is the moment of time symmetry $t = 0$, where the S^3 has its minimum radius L (the universe contracts for $t < 0$ and expands for $t > 0$).



The dashed diagonals are the horizons of the observer on the left edge, whose worldline ends at the top-left corner. The diagonal from top-left to bottom-right is the boundary of the observer's causal past, the *event horizon*: signals emitted from the region above it (labelled "hidden") can never reach the observer, and it is also the limit of the observer's past light cone (the particle horizon) as $t \to \infty$; at any finite time the observer sees only a smaller region (the red past light cone is an example). The diagonal from bottom-left to top-right is the boundary of the causal future of the observer's worldline: the observer can never influence events below it. A light ray from one pole reaches the opposite pole only at a corner of the diagram, i.e. at $\mathcal{I}^\pm$. Hence the de Sitter Penrose diagram is a square with spacelike $\mathcal{I}^\pm$ and horizons for every geodesic observer.

Question 5. *(6 points)*

a) Linearised diffeomorphisms $x^\mu \to x^\mu + \xi^\mu$ act as $h_{\mu\nu} \to h_{\mu\nu} - \partial_\mu \xi_\nu - \partial_\nu \xi_\mu$, so

$$\bar{h}_{\mu\nu} \to \bar{h}_{\mu\nu} - \partial_\mu \xi_\nu - \partial_\nu \xi_\mu + \eta_{\mu\nu} \partial_\rho \xi^\rho \,, \qquad \partial^\mu \bar{h}_{\mu\nu} \to \partial^\mu \bar{h}_{\mu\nu} - \partial^\rho \partial_\rho \xi_\nu \,. \tag{17}$$

Choosing ξ_ν to solve $\partial^\rho \partial_\rho \xi_\nu = \partial^\mu \bar{h}_{\mu\nu}$ (always possible: this is a wave equation with a source) imposes the Lorenz (de Donder) gauge $\partial^\mu \bar{h}_{\mu\nu} = 0$. Then the second and third terms of the equation vanish and

$$-\frac{1}{2}\, \partial^\rho \partial_\rho\, \bar{h}_{\mu\nu} = 8\pi G\, T_{\mu\nu} \quad \Longleftrightarrow \quad \left(-\partial_t^2 + \nabla^2\right) \bar{h}_{\mu\nu} = -16\pi G\, T_{\mu\nu} \,. \tag{18}$$

In vacuum this is the wave equation $\Box \bar{h}_{\mu\nu} = 0$: the perturbations propagate at the speed of light as *gravitational waves*, $\bar{h}_{\mu\nu} = A_{\mu\nu} e^{ik\cdot x}$ with $k^\mu k_\mu = 0$ and $k^\mu A_{\mu\nu} = 0$.

(The residual gauge freedom $\Box\xi = 0$ can be used to reach the transverse-traceless gauge, leaving two physical polarisations, + and $\times$.)

b) The perturbation has $h_{0\mu} = 0$ (TT gauge). The geodesic equation for a particle initially at rest, $u^\mu = (1, 0, 0, 0)$, has

$$\frac{d^2x^i}{d\tau^2} = -\Gamma^i_{00} = -\frac{1}{2}\eta^{ij}\left(2\partial_0 h_{j0} - \partial_j h_{00}\right) = 0\,. \tag{19}$$

Hence the free particles stay at *fixed coordinate positions*: coordinates are comoving with the test masses, and the wave shows up only as a change in the metric components.

At fixed time t the spatial metric is $g_{xx} = 1+h_+(t-z)$, $g_{yy} = 1-h_+(t-z)$, $g_{zz} = 1$ (and $|h_+| \ll 1$). The two particles on the x-axis both have $y = z = 0$, so $h_+(t-z) = h_+(t)$ everywhere on the segment joining them, and

$$\Delta s = \int_0^{L_x} \sqrt{g_{xx}}\,dx = \int_0^{L_x} \sqrt{1+h_+(t)}\,dx = L_x\sqrt{1+h_+(t)} \approx L_x\left(1 + \frac{1}{2}h_+(t)\right). \tag{20}$$

For the third particle at $(0, L_y, 0)$, the segment lies along y at $x = z = 0$:

$$\Delta s = \int_0^{L_y} \sqrt{g_{yy}}\,dy = L_y\sqrt{1-h_+(t)} \approx L_y\left(1 - \frac{1}{2}h_+(t)\right). \tag{21}$$

So the wave alternately stretches the x-separation and squeezes the y-separation, and vice versa half a period later, with a *differential* strain $(\Delta s_x/L_x) - (\Delta s_y/L_y) = h_+(t)$.

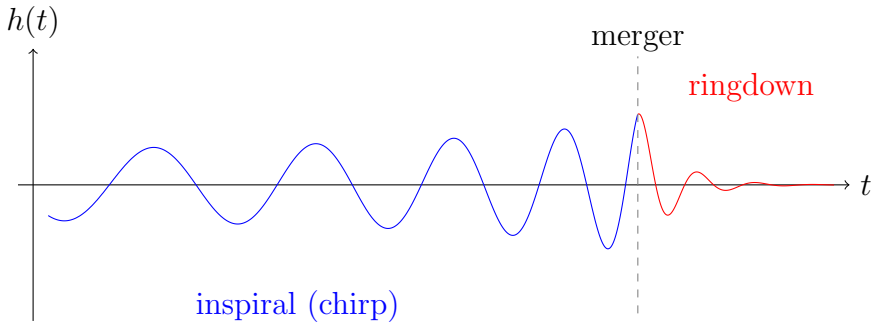
c) **Qualitative picture.** Two masses m_1, m_2 in orbit have a time-dependent mass quadrupole moment, which radiates gravitational waves (there is no monopole or dipole radiation by conservation of mass and momentum). The waves have frequency twice the orbital frequency, $f_{\rm GW} = 2f_{\rm orb}$, and carry away energy and angular momentum. The orbit therefore shrinks, the bodies orbit faster, and the radiation becomes stronger—a runaway process that ends in a merger. (This was confirmed indirectly by the decay of the Hulse–Taylor binary pulsar orbit.) Three phases:

- *Inspiral:* the separation shrinks slowly and adiabatically, quasi-circular orbits with increasing frequency and amplitude, so the signal is a *chirp*. At leading order

$$f_{\rm GW} \propto (t_c - t)^{-3/8}, \qquad h \propto \frac{(G\mathcal{M}_c)^{5/3} f^{2/3}}{c^4\, d}\,, \qquad \mathcal{M}_c = \frac{(m_1 m_2)^{3/5}}{(m_1+m_2)^{1/5}}\,, \tag{22}$$

  where d is the distance to the source and t_c the coalescence time. The rate at which the frequency sweeps up measures the chirp mass $\mathcal{M}_c$, the amplitude then gives the distance d, and subtle higher-order effects encode the mass ratio and spins.

- *Merger:* when the separation reaches roughly the innermost stable circular orbit, $r \sim 6GM/c^2$ (at $f_{\rm GW} \sim c^3/(6^{3/2}\pi GM) \approx 4.4\,{\rm kHz}\,(M_\odot/M)$), the bodies plunge together and merge into a single, highly distorted black hole. The signal reaches maximum amplitude and frequency; the dynamics is strongly non-linear and strongly curved, and is accessible only through numerical relativity. A few per cent of the total rest mass is radiated in the last few cycles (about $3M_\odot c^2$ in GW150914).
- *Ringdown:* the remnant black hole radiates away its deformations as a superposition of damped sinusoids (quasi-normal modes), $h \sim e^{-t/\tau}\cos(2\pi f_{\rm qnm} t)$, and settles to a stationary Kerr black hole. The frequency and decay time depend only on the final mass and spin (no-hair theorem), so the ringdown tests general relativity and measures the remnant's properties.



The amplitude and frequency both rise through the inspiral, peak at merger, and then the signal decays exponentially at a nearly constant frequency.

Interferometric detection. A detector such as LIGO is a Michelson interferometer with two perpendicular arms, each of length $L \approx 4\,{\rm km}$ (made effectively longer, $\sim 1000\,{\rm km}$, by Fabry–Pérot cavities), with suspended mirrors acting as free test masses (they are free to move horizontally for frequencies above the pendulum frequency). A laser beam is split at a beam splitter, travels down both arms, reflects, and recombines at a photodetector; the arm lengths are tuned so that, without a wave, the light interferes destructively at the output (a *dark fringe*). By part b), a + polarised wave incident from above, with the arms along x and y, stretches one arm and shrinks the other: $\Delta L_x = +\frac{1}{2}Lh_+$, $\Delta L_y = -\frac{1}{2}Lh_+$, so the *differential* arm-length change is $\Delta L_x - \Delta L_y = L\,h_+(t)$. This changes the round-trip light travel times and so the relative phase of the two beams, $\Delta\phi \simeq 4\pi L\,h_+/\lambda$ (for a single round trip), and the light power at the photodetector varies in proportion to $h(t)$. The detector therefore records the wave strain directly as a time series: $h \sim 10^{-21}$ means $\Delta L \sim 10^{-18}\,{\rm m}$, far smaller than a proton, and this requires great isolation from noise. (A $\times$ polarised wave stretches along the diagonals; arbitrary sources give $h = F_+h_+ + F_\times h_\times$ with antenna factors $F_{+,\times}$, so several detectors are needed to localise a source.)

Connecting signal to physics. Frequency evolution during inspiral gives $\mathcal{M}_c$; the amplitude then gives the distance; the merger frequency, $\propto 1/M$, gives the total mass; the ringdown frequency and damping time give the final mass and spin; the ratio of the $+$ and $\times$ amplitudes tells the inclination of the orbital plane; and the total radiated energy is the difference between the initial total mass and the final mass.